

DES Instrument Cluster

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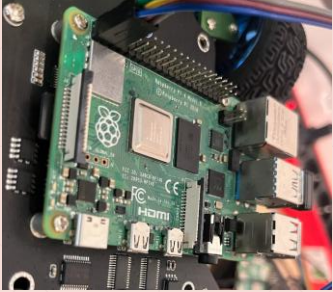
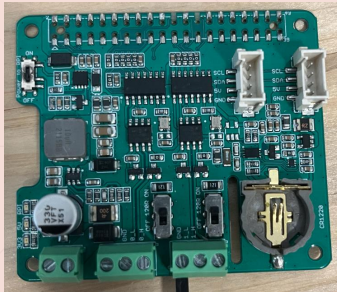
- Specification
- GUI
- Display
- Flow
- CAN Communication
- D-Bus
- Cross Compile

Display

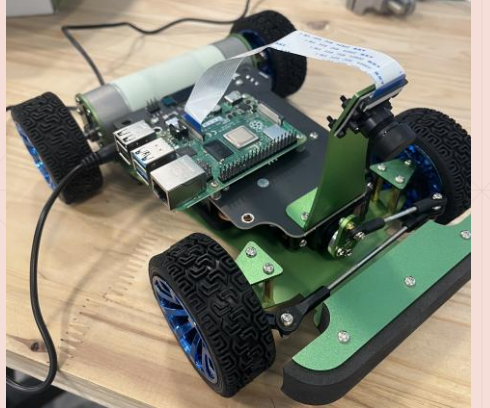


Specification : Hardware

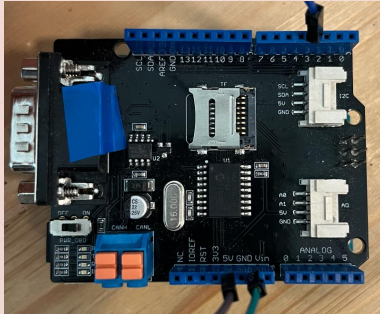
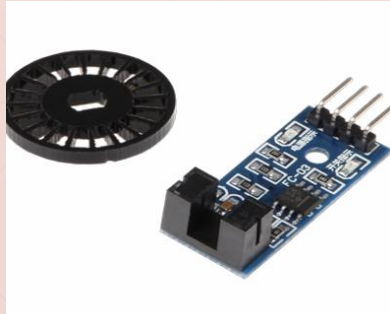
Raspberry Pi

Raspberry Pi 4	CAN_BUS(FD) Shield(MCP2518FD)	7.9-inch Display
		

Piracer

Piracer Kit


Arduino UNO

Arduino UNO	CAN-BUS Shield (MCP2515)	Speed Sensor (LM 393)
		

Joystick

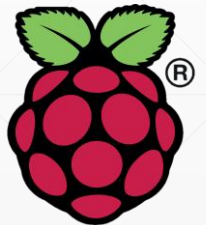


Specification : Software

- Languages : C++, Python, QML

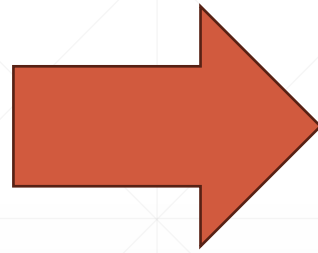


- Frameworks / Libraries : Qt5, dbus-python, piracer gamepads, PyGObject
- IPC (Inter-Process Communication) : D-Bus
- Build System : Qt-based Cross-Compilation (PC -> Raspberry Pi)

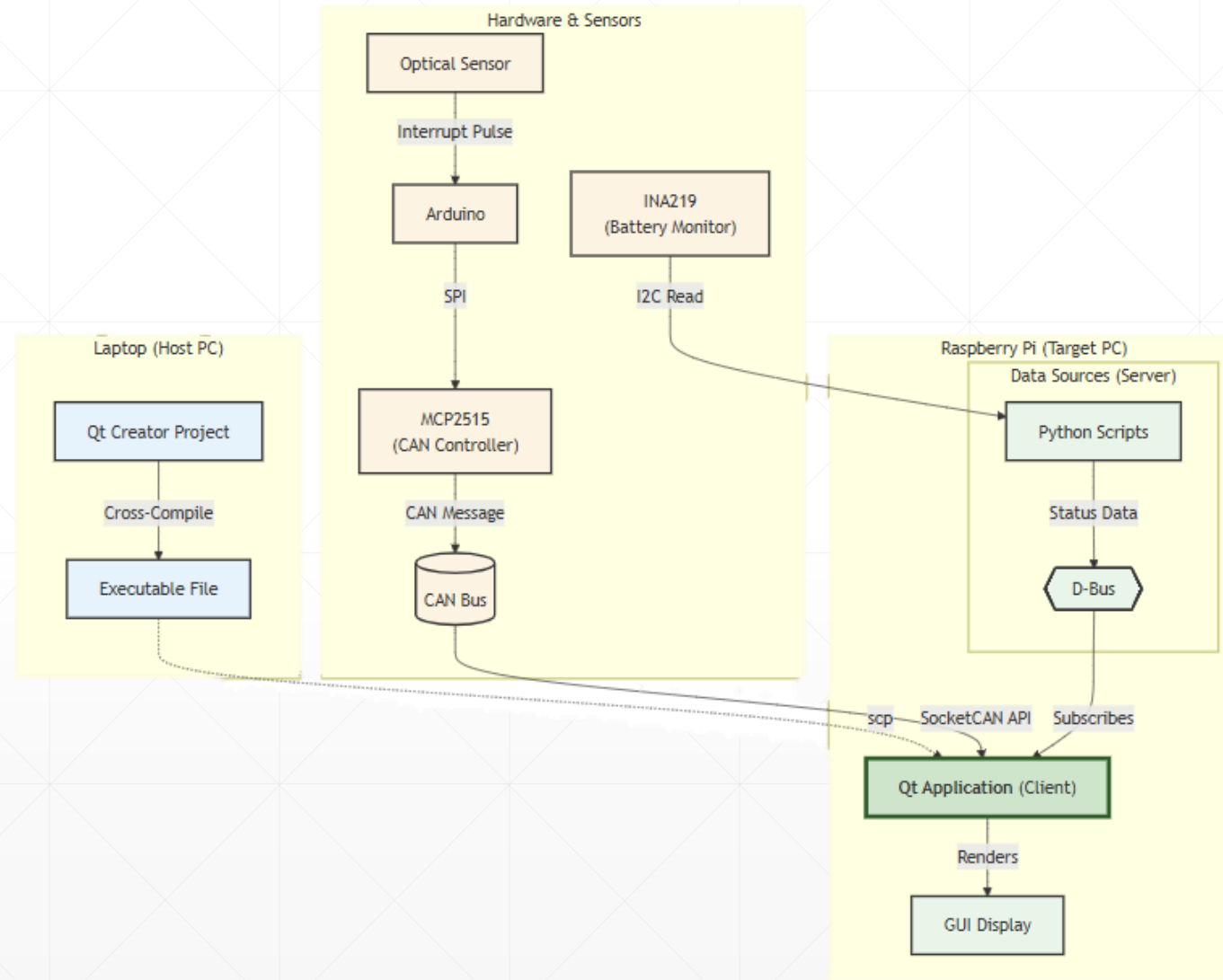


Raspberry Pi

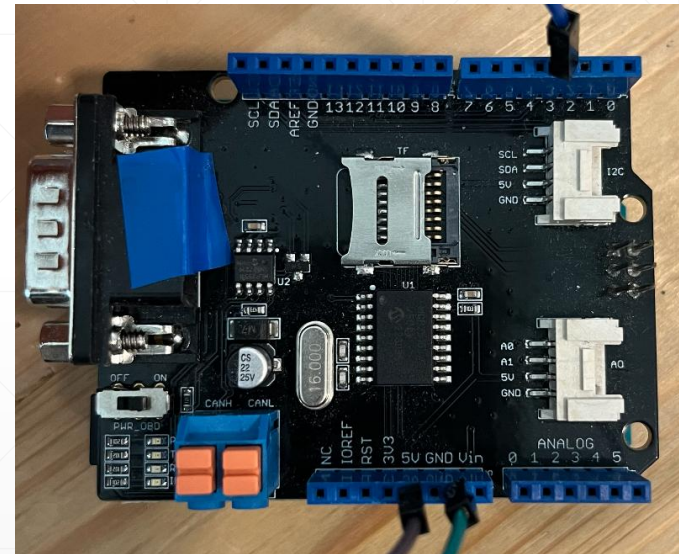
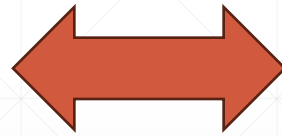
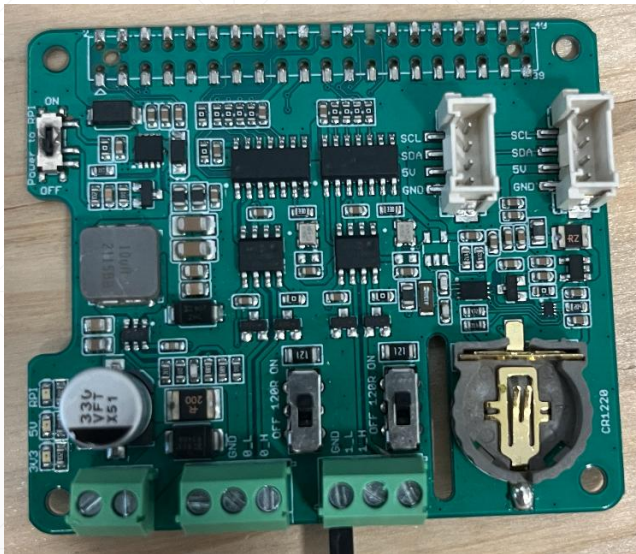
GUI



Flow



CAN Communication



D-Bus



-> Sender (soc.py, vehicle_controller.py)

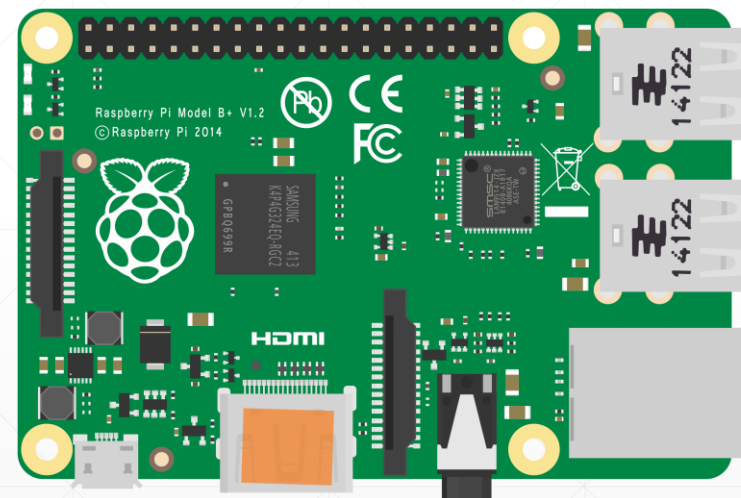


-> Receiver (dbusreceiver.cpp)

Cross Compile



.exe



Thank you

Q&A